

PATUXENT RIVER NAS, MD, USA

WMO: 724040

Lat: 38.300N

Lon: 76.417W

Elev: 39

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13721

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.0	20.7	-0.7	5.3	20.3	3.8	6.7	24.1	28.9	29.5	26.7	29.9	11.2	330	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	92.7	76.4	90.1	75.7	87.8	74.9	79.4	88.2	78.0	86.4	76.8	84.8	8.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.0	141.0	83.8	75.4	133.1	82.4	74.3	128.5	81.5	42.8	88.3	41.4	86.3	40.2	84.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.5	20.5	18.3	DB	10.7	97.6	5.1	2.9	7.0	99.7	4.0	101.4	1.1	103.1	-2.6	105.2
			WB	8.8	82.5	4.7	2.2	5.4	84.1	2.7	85.5	0.0	86.7	-3.4	88.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.0	36.2	38.8	45.6	56.5	66.0	74.6	78.6	76.9	70.9	60.3	49.0	41.8
	DBStd	16.71	10.19	9.30	9.20	8.50	7.58	5.89	4.85	4.69	5.97	7.82	7.96	8.94
	HDD50	1408	443	332	198	29	0	0	0	0	0	8	112	286
	HDD65	4006	892	734	604	282	83	5	0	0	16	190	481	719
	CDD50	4343	16	19	62	222	496	739	886	833	628	326	83	33
	CDD65	1467	0	0	3	26	114	294	421	368	194	43	3	1
	CDH74	12269	3	1	25	192	809	2426	4102	3231	1244	229	5	2
	CDH80	4048	0	0	4	46	226	810	1577	1045	299	41	0	0
	WSAvg	8.4	9.3	9.3	9.9	9.5	8.4	7.7	7.0	6.7	7.6	8.2	8.7	8.5
	PrecAvg	40.7	3.3	3.1	3.9	2.9	4.1	3.7	3.7	3.8	3.5	3.0	2.8	3.3
Precipitation	PrecMax	60.8	6.9	7.0	10.2	8.2	11.3	7.4	10.5	9.4	8.0	6.6	6.0	7.1
	PrecMin	28.3	0.4	0.8	0.8	0.4	0.5	0.9	0.8	0.6	0.1	0.0	0.3	0.5
	PrecStd	9.3	1.5	1.5	2.2	1.6	2.8	2.0	2.4	2.4	2.2	1.7	1.6	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.0	70.2	77.5	85.4	90.3	94.2	98.0	94.7	91.4	84.2	73.5	70.4
		MCWB	59.8	59.4	60.7	65.0	73.7	75.4	77.7	77.6	75.0	72.4	64.2	62.2
	2%	DB	62.6	63.5	70.3	79.4	85.7	90.5	93.3	90.8	86.4	79.6	69.2	65.3
		MCWB	56.6	55.1	58.6	64.5	71.8	74.8	76.9	76.6	73.6	69.5	62.0	59.3
	5%	DB	57.1	57.4	65.2	74.6	81.9	87.8	90.3	88.2	83.3	76.0	65.6	60.7
		MCWB	51.6	50.1	55.9	63.0	70.2	73.9	76.4	75.8	72.4	67.9	59.5	56.0
	10%	DB	51.7	52.5	60.7	70.5	78.7	84.6	87.9	85.6	80.7	72.6	62.1	55.4
		MCWB	46.5	46.2	52.6	60.6	68.4	72.9	75.6	74.4	71.3	65.2	56.2	50.3
	0.4%	WB	61.8	62.7	64.4	70.4	76.0	79.1	81.8	81.2	78.5	75.0	66.8	65.4
		MCDB	65.9	68.1	73.6	78.1	86.6	88.3	89.6	88.0	85.8	80.1	71.1	69.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	57.5	57.1	60.8	67.2	73.5	77.4	79.9	78.9	76.3	72.2	64.0	60.3
		MCDB	62.0	62.3	68.3	75.4	82.0	85.8	88.4	87.6	82.5	78.3	67.7	64.3
	5%	WB	52.8	51.8	57.3	64.6	71.8	76.3	78.3	77.6	74.9	69.4	60.3	56.4
		MCDB	56.4	56.4	63.9	72.2	79.8	84.3	87.1	85.4	80.7	74.5	64.3	60.3
	10%	WB	46.8	46.9	53.4	62.2	69.8	75.0	77.0	76.3	73.3	66.4	57.1	51.3
		MCDB	50.6	51.8	59.2	69.7	77.0	82.5	85.6	83.5	78.8	71.3	61.2	55.0
Mean Daily Temperature Range	5% DB	MDBR	14.7	15.7	16.4	17.7	16.4	15.8	15.5	15.0	15.1	16.5	16.1	15.1
		MCDBR	21.7	22.8	23.7	22.8	19.9	18.9	18.4	16.9	17.5	18.7	19.9	20.4
		MCWBR	17.7	17.4	15.2	12.4	10.2	8.2	7.5	7.5	8.4	11.0	13.9	16.4
	5% WB	MCDBR	20.9	21.2	21.4	19.9	17.9	16.6	16.1	14.9	14.3	16.0	16.6	18.9
		MCWBR	18.3	17.9	16.0	12.7	9.8	8.2	7.9	7.5	8.1	10.7	13.8	16.5
Clear-Sky Solar Irradiance	taub		0.311	0.319	0.354	0.393	0.434	0.485	0.516	0.507	0.410	0.360	0.326	0.311
	taud		2.511	2.479	2.426	2.332	2.250	2.157	2.084	2.093	2.341	2.485	2.539	2.546
	Ebn at Noon		277	289	288	281	270	256	246	245	265	271	269	269
	Edh at Noon		25	29	34	39	43	48	51	49	36	29	24	23
All-Sky Solar Radiation	RadAvg		699	959	1280	1616	1839	2020	1959	1755	1420	1081	790	615
	RadStd		53	102	107	115	204	150	94	96	143	118	68	54

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (79 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+254	+157

Nomenclature: See separate page